

2021 10%: selected excess 2 at a 79 MeV signal hypothesis

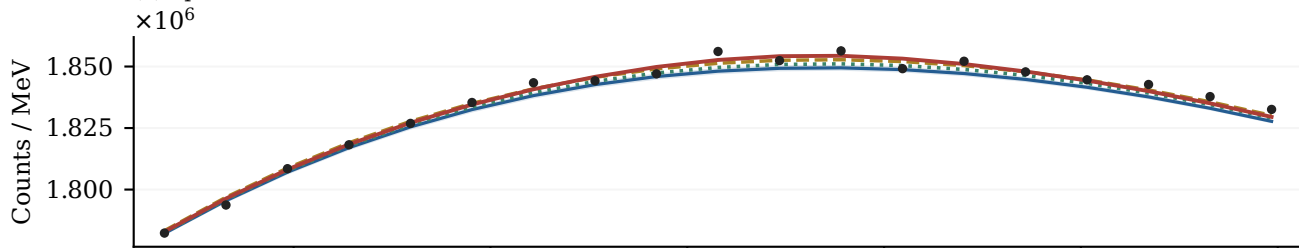
v16 neighboring signal-MC template | $c = 77.137$ MeV, $s = 1.633$ MeV

Signed full-selected yield = $24,979 \pm 9,927$; local asymptotic $Z = 2.52$ σ , $p = 0.00592$

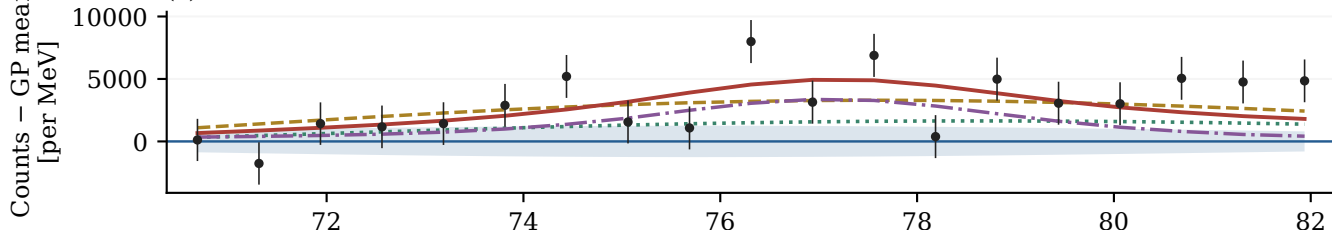
GP background-only toys: local excess rank $p = 0.022$ (21 / 1000 more extreme toys); count 95% interval [0.013, 0.0319]

- Observed counts
- ⋯ Background in signed S+B fit
- ⬜ Blind region: excluded from GP training
- ⬜ GP mean and ± 1 SD constraint
- Signed signal + background
- Search mass hypothesis
- Background-only fit
- Signed signal (middle panel)
- Signal core center

(a) Spectrum in the fit interval



(b) Residuals relative to the GP mean



(c) Standardized residuals and blind window

